Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths *	Prac **
6	*(a)*	(i)		Lay out grid/ lay out two tapes (1) Random number {table/generator}/ use (10 sided) dice} to give coordinates (1) Avoids bias/ better representation of the whole area/ avoid user generated influence (1)	3			3		3
		(ii)		Secondary succession (1) colonised prior to being cleared, so {seeds/nutrients/ humus} already present(1) (secondary succession) changes environment and allows other species to grow (1)	2	1		3		
		(iii)		Increased (1) More habitats/ food sources/ niches (1)		2		2		
	(b)	(i)		Carry out nitrogen fixation (1) Nitrogen gas converted to nitrogen containing compounds/ ammonium ions(1) NOT nitrate Used by plants to make amino acids/proteins/DNA (1)	2	1		3		
		(ii)		A. (Leghaemoglobin) {combines/ binds} with the oxygen present (1) NOT leghaemoglobin absorbs oxygen B. Leghaemoglobin gives {anaerobic/ oxygen free/ low oxygen} conditions (in the root nodule)(1) C. *Rhizobium* only fixes nitrogen in {anaerobic/ oxygen free/ low oxygen } conditions(1)			3			

Question				Marking Details	Marks Available					
					AO1	AO2	AO3	Total	Maths *	Prac **
	(c)	(i)		stable community/no further succession/final community/ community has reached equilibrium (1)	1			1		
		(ii)		{Waterlogged soils/ conditions} {lack oxygen/ are anaerobic}(1) roots unable to respire aerobically/ active transport of minerals stops(1) ***OR*** Denitrification (1) lack of nitrate for trees (1)	1	1		2		
	(d)			Any three from: A. Nitrates are in low supply (1) B. It takes a lot of ATP to {absorb nitrates/fix nitrogen/ change nitrogen to ammonia} (1) C. More energy efficient to digest insects (than to fix nitrates) (1) D. Used as a N source from amino acids/protein/ nucleic acids; (1)			3	3		
				Question 6 total	**9**	**5**	**6**	**17**		**3**